Remaining Steps in the Level-Set Rearrangement Route

Same-Centre Fusco–Julin, Metric First Variation, and a Bad-Density Repair

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Abstract

This note records the outcome of attacking the three remaining gaps isolated in the debug note for the continuous-rearrangement proof. The first gap, the same-centre Fusco–Julin input, is actually closed: the correct import is not the separately minimised statement currently written in `preliminaries_quant_isoperimetry.tex`, but the centred consequence of Fusco–Julin's oscillation index together with their Proposition 1.2 (equivalently Proposition 5.5 in Fusco's survey). The second gap, the finite-perimeter metric first variation for the rescaled curve $\rho^{-1}E_\rho$, is still not proved by the current manuscript: the Reshetnyak/AGS closure has the lower-semicontinuity inequality in the wrong direction. The useful new point is that the third gap, the bad-density kinetic estimate, can be bypassed if the trace is run on the unscaled distance

$$q(\rho) := \inf_z |E_\rho \Delta B_\rho(z)|$$

to the moving ball. Nestedness gives an unconditional Lipschitz bound for q, so the bad-density set costs $O(\delta_T)$. The remaining local theorem is then a raw moving-ball first-variation estimate on good levels; in the smooth case it is exactly the vector-field computation below, and in finite perimeter it is a limsup deformation theorem, strictly weaker than the old rescaled metric theorem.

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1 Setup

Normalize $|\Omega| = \omega_n$, $\Omega \subset B_R$, and let u be the torsion function. Write

$$E_\rho := \{u > t(\rho)\}, \qquad |E_\rho| = \omega_n \rho^n, \qquad w(\rho) := -t'(\rho), \qquad d\mu = w(\rho)\, d\rho.$$

The level-set identity gives, on every fixed bulk interval $[\rho_*, \rho_\delta] \subset (0,1]$,

$$\int_{\rho_*}^{\rho_\delta} \big(D_H(t(\rho)) + D_I(t(\rho))\big)\, d\mu(\rho) \le C(n, \rho_*)\, \delta_T(\Omega). \tag{1}$$

Here D_H is the Cauchy–Schwarz defect of $|\nabla u|$ on $\partial^* E_\rho$, and

$$D_I(t(\rho)) = P(E_\rho)^2 - P(B_\rho)^2.$$

Talenti's profile-gap estimate gives the bad-density bound

$$|B_\tau| \le C(n, \rho_*)\, \delta_T(\Omega), \qquad B_\tau := \{\rho \in [\rho_*, \rho_\delta] : w(\rho) < \tau_0\}, \quad \tau_0 := \frac{\rho_*}{2n}. \tag{2}$$

2 Same-Centre Fusco–Julin Is Closed

2.1 Where the current repository proof goes wrong

The proof of the scaled same-centre theorem in `Manuscript/preliminaries_quant_isoperimetry.tex` currently chooses an oscillation-minimising centre z and then uses

$$\frac{|E \Delta B_\rho(z)|}{\omega_n \rho^n} \ge \mathcal{A}(E)$$

as if it were the needed upper bound. This is the precise sign error: the Fraenkel asymmetry is an infimum over centres, so it gives only a lower bound on the centred symmetric difference at a prescribed centre. That proof cannot be patched by algebra.

The statement, however, is available from the correct Fusco–Julin input. Fusco–Julin do not merely give two unrelated minimisations. Their oscillation index is minimised over balls, and their Poincare-type estimate shows that the same ball controls the L^1 displacement.

2.2 Import-ready statement

For $|E| = |B_\rho|$, define the pointed oscillation

$$\beta(E, z)^2 := \frac{1}{2} \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1}(x).$$

Theorem 2.1 (Same-centre Fusco–Julin package)**.** *Let $n \ge 2$ and $0 < \rho_* \le \rho \le 1$. There is a constant $C_{\mathrm{scFJ}} = C_{\mathrm{scFJ}}(n, \rho_*)$, explicitly traceable to the Fusco–Julin constants, with the following property. If E is a finite-perimeter set with $|E| = \omega_n \rho^n$, then there is a centre z_E such that*

$$|E \Delta B_\rho(z_E)|^2 + \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z_E}{|x - z_E|} \right|^2 d\mathcal{H}^{n-1}(x) \le C_{\mathrm{scFJ}} \big(P(E) - P(B_\rho)\big). \tag{3}$$

Consequently, on a level E_ρ,

$$|E_\rho \Delta B_\rho(z_\rho)|^2 + \int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1} \le C(n, \rho_*) D_I(t(\rho)). \tag{4}$$

Proof. After translating and scaling, it is enough to treat $\rho = 1$. Let z_E be a minimiser of the Fusco–Julin oscillation index. Fusco–Julin's strong form gives

$$\beta(E, z_E)^2 \le C_{FJ}(n)\big(P(E) - P(B_1)\big).$$

Their Poincare-type estimate for the same minimising ball gives, at this same centre,

$$|E \Delta B_1(z_E)|^2 \le C(n)\, \beta(E, z_E)^2.$$

Equivalently, in the notation of [1, Prop. 1.2] or [2, Prop. 5.5], once the oscillation centre has been translated to the origin,

$$\beta(E,0)^2 \geq c(n)|E\Delta B_1|^2$$

up to the harmless perimeter-deficit term that is already dominated by the same strong inequality. Combining the two estimates proves (3) for $\rho = 1$.

For general ρ, apply the unit-radius statement to $\widetilde{E} := \rho^{-1}(E - z_E)$. The oscillation integral scales like ρ^{1-n}, the perimeter deficit like ρ^{1-n}, and $|E\Delta B_\rho(z_E)|^2 = \rho^{2n}|\widetilde{E}\Delta B_1|^2$. Since $\rho \geq \rho_*$, all scale factors are absorbed into $C_{\mathrm{scFJ}}(n, \rho_*)$. Finally,

$$P(E_\rho) - P(B_\rho) = \frac{D_I(t(\rho))}{P(E_\rho) + P(B_\rho)} \leq C(n, \rho_*) D_I(t(\rho)),$$

which gives (4). □

Remark 2.2 (Effectivity of the constant)**.** The argument above introduces no compactness beyond the constant already present in the Fusco–Julin theorem. Thus the same-centre repair is quantitative in exactly the same sense as the imported Fusco–Julin inequality: once the constants in Theorem 5.4 and Proposition 5.5 of the source are unpacked, C_{scFJ} is obtained by algebra and scaling.

Remark 2.3. This also repairs the measurable centre selection: choose the minimisers of the joint closed sublevel set supplied by (3). The selection is standard once the centre is restricted to a compact ball; compactness follows from $E_\rho \subset B_R$ and the centred L^1 bound.

3 The Current Rescaled Metric Closure Is Not Sound

The manuscript file `Manuscript/plan2_metric_framework.tex` tries to pass from smooth approximations to finite-perimeter level sets by combining Reshetnyak lower semicontinuity with AGS lower semicontinuity of metric derivatives. The decisive lines have the following form. For smooth approximants,

$$|\dot{F}^k_\rho|_{\mathcal{X}} \leq m_k(\rho),$$

and weak compactness gives $m_k \rightharpoonup m$. Reshetnyak gives only

$$m(\rho) \geq \rho_*^{-n} \inf_a \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a \cdot \nu_\rho| \, d\mathcal{H}^{n-1}.$$

AGS then gives $|\dot{F}_\rho|_{\mathcal{X}} \leq m(\rho)$. These two inequalities do not imply the desired estimate by the boundary action; one would need $m(\rho) \leq$ that action, or a direct limsup recovery. Thus the old rescaled finite-perimeter metric theorem remains open.

4 A Bad-Density Bypass: Use the Moving Ball

The bad-density obstruction was caused by the rescaled curve

$$F_\rho = \rho^{-1} E_\rho.$$

The rescaling differentiates rough sets, and boundedness plus nestedness do not imply a perimeter-free Lipschitz bound for $\rho \mapsto \rho^{-1}E_\rho$. The following scalar quantity avoids that operation:

$$q(\rho) := \inf_{z \in \mathbb{R}^n} |E_\rho \Delta B_\rho(z)|. \tag{5}$$

Lemma 4.1 (Unconditional Lipschitz bound for q). *For $0 < \rho \le \sigma \le 1$,*

$$|q(\sigma) - q(\rho)| \le 2\omega_n(\sigma^n - \rho^n) \le 2n\omega_n|\sigma - \rho|.$$

In particular $q \in W^{1,\infty}_{\mathrm{loc}}((0,1))$, with a computable constant depending only on n.

Proof. Fix z with $|E_\rho \Delta B_\rho(z)| \le q(\rho) + \varepsilon$. Since the level sets are nested and the same-centre balls are nested,

$$\begin{aligned}|E_\sigma \Delta B_\sigma(z)| &\le |E_\rho \Delta B_\rho(z)| + |E_\sigma \setminus E_\rho| + |B_\sigma(z) \setminus B_\rho(z)| \\ &= |E_\rho \Delta B_\rho(z)| + 2\omega_n(\sigma^n - \rho^n).\end{aligned}$$

Taking the infimum in z and then $\varepsilon \downarrow 0$ gives

$$q(\sigma) \le q(\rho) + 2\omega_n(\sigma^n - \rho^n).$$

Interchanging ρ and σ gives the reverse inequality. □

Corollary 4.2 (Bad-density action is free for q). *With B_τ as in* (2),

$$\int_{B_\tau} |q'(\rho)|^2 \, d\rho \le C(n, \rho_*) \, \delta_T(\Omega).$$

Proof. By Lemma 4.1, $|q'| \le 2n\omega_n$ a.e. Combine this with (2). □

5 The Scalar Trace Lemma

The moving-ball distance gives the endpoint extraction without any rescaled bad-level kinetic estimate.

Lemma 5.1 (Scalar moving-ball trace). *Let $J = [a, b] \subset [\rho_*, \rho_\delta]$ have length bounded below by $\ell_0 > 0$. Assume*

$$\int_J q(\rho)^2 \, d\mu(\rho) \le C_0 \delta_T, \tag{6}$$

$$\int_{J \cap G_\tau} |q'(\rho)|^2 \, d\rho \le C_1 \delta_T, \qquad G_\tau := J \setminus B_\tau, \tag{7}$$

$$|B_\tau \cap J| \le C_2 \delta_T. \tag{8}$$

Then, for δ_T small enough depending on n, ρ_*, ℓ_0,

$$\sup_{\rho \in J} q(\rho)^2 \le C(n, \rho_*, \ell_0, C_0, C_1, C_2) \, \delta_T.$$

Proof. First convert the weighted L^2 distance into a Lebesgue L^2 distance. On G_τ, $d\mu \ge \tau_0 \, d\rho$, hence

$$\int_{J \cap G_\tau} q^2 \, d\rho \le \tau_0^{-1} C_0 \delta_T.$$

On B_τ, the trivial bound $q(\rho) \le |E_\rho| + |B_\rho| = 2\omega_n \rho^n \le 2\omega_n$ and (8) give

$$\int_{J \cap B_\tau} q^2 \, d\rho \le 4\omega_n^2 C_2 \delta_T.$$

Thus $\int_J q^2 \, d\rho \le C\delta_T$, and there is $\rho_0 \in J$ with

$$q(\rho_0)^2 \le \frac{C}{|J|} \delta_T \le C(\ell_0) \delta_T.$$

Next,

$$\int_J |q'|^2\,d\rho = \int_{J\cap G_\tau} |q'|^2\,d\rho + \int_{J\cap B_\tau} |q'|^2\,d\rho \le C\delta_T$$

by (7) and Corollary 4.2. For every $\rho \in J$,

$$q(\rho) \le q(\rho_0) + \int_J |q'| \le C\delta_T^{1/2} + |J|^{1/2}\left(\int_J |q'|^2\right)^{1/2} \le C\delta_T^{1/2}.$$

Squaring proves the claim. □

6 The Remaining Local Theorem

The preceding sections reduce the metric part to a local good-level estimate for q. This theorem is the precise replacement for the old rescaled finite-perimeter metric theorem.

Problem 6.1 (Raw moving-ball first variation)**.** Let $E_\rho = \{u > t(\rho)\}$ be a.e. finite-perimeter levels. On the good set where $D_I(t(\rho))$ is small, choose the same-centre Fusco–Julin centre z_ρ from Theorem 2.1. Prove the finite-perimeter limsup estimate

$$|q'(\rho)| \le C(n,R,\rho_*)q(\rho) + C(n,R,\rho_*)\inf_{a\in\mathbb{R}^n}\int_{\partial^* E_\rho} |V_\rho - e_{z_\rho}\cdot\nu_\rho - a\cdot\nu_\rho|\,d\mathcal{H}^{n-1} \tag{9}$$

for a.e. good ρ, where

$$V_\rho = \frac{w(\rho)}{|\nabla u|}, \qquad e_{z_\rho}(x) = \frac{x - z_\rho}{|x - z_\rho|}.$$

Proposition 6.2 (Why (9) closes the route)**.** *If Problem 6.1 is proved, then the bounded sharp Saint–Venant estimate*

$$\mathcal{A}(\Omega)^2 \le C(n,R)\delta_T(\Omega)$$

follows from the level-set identity and the boundary-layer transfer, with computable constants.

Proof. On a good level, Theorem 2.1 gives

$$q(\rho)^2 \le CD_I(t(\rho)), \qquad \int_{\partial^* E_\rho} |1 - e_{z_\rho}\cdot\nu_\rho|\,d\mathcal{H}^{n-1} \le CD_I(t(\rho)).$$

The velocity variance identity gives

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar V_\rho|\,d\mathcal{H}^{n-1}\right)^2 \le CD_H(t(\rho)), \qquad \bar V_\rho := \frac{P(B_\rho)}{P(E_\rho)}.$$

Since $P(E_\rho) - P(B_\rho) \le CD_I(t(\rho))$ on the good set,

$$\begin{aligned}\int_{\partial^* E_\rho} |V_\rho - e_{z_\rho}\cdot\nu_\rho| &\le \int |V_\rho - \bar V_\rho| + P(E_\rho)|\bar V_\rho - 1| + \int |1 - e_{z_\rho}\cdot\nu_\rho| \\ &\le CD_H(t(\rho))^{1/2} + CD_I(t(\rho)).\end{aligned}$$

Squaring and using $D_I \le \eta_I$ on the good set,

$$|q'(\rho)|^2 \le C\big(D_H(t(\rho)) + D_I(t(\rho))\big)$$

there. On G_τ, $d\rho \le \tau_0^{-1}d\mu$, so

$$\int_{G_\tau} |q'|^2\,d\rho \le C\int_{\rho_*}^{\rho_\delta} (D_H + D_I)\,d\mu \le C\delta_T$$

by (1). The distance hypothesis $\int q^2\,d\mu \le C\delta_T$ follows from $q^2 \le CD_I$ on the good isoperimetric set and the usual bad-I weighted Markov estimate with the trivial bound $q \le 2\omega_n$. Lemma 5.1 gives

$$q(\rho_\delta)^2 \le C\delta_T$$

on an order-one window ending at ρ_δ (or at the good endpoint $\widehat{\rho}$). Therefore

$$\mathcal{A}(E_{\rho_\delta})^2 \le C(n,\rho_*)q(\rho_\delta)^2 \le C\delta_T.$$

Finally $E_{\rho_\delta} \subset \Omega$ and

$$|\Omega \setminus E_{\rho_\delta}| = \omega_n(1-\rho_\delta^n) \le C(n)\sqrt{\delta_T},$$

so the elementary boundary-layer transfer gives

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_{\rho_\delta}) + C\sqrt{\delta_T},$$

and the desired squared estimate follows. □

6.1 Smooth proof of the local theorem

For smooth moving level surfaces, Problem 6.1 is exactly the standard relative first variation. Fix a centre z and a translation velocity a. Let $X = e_z + a$ near $\partial B_\rho(z)$, with a harmless smooth cutoff away from that sphere so that the flow is globally Lipschitz. The flow of $B_\rho(z)$ has boundary normal velocity $1 + a\cdot\nu$, hence it agrees to first order with $B_{\rho+h}(z+ha)$. The level surface ∂E_ρ has normal velocity V_ρ. Differentiating $|E_\rho \Delta B_\rho(z)|$ under these two motions gives

$$\frac{d^+}{d\rho}|E_\rho \Delta B_\rho(z)| \le Cq(\rho) + \int_{\partial E_\rho} |V_\rho - e_z\cdot\nu_\rho - a\cdot\nu_\rho|\,d\mathcal{H}^{n-1}.$$

The Cq term is the Jacobian error on the already mismatched region and is absorbed by $q^2 \le CD_I$. Taking the infimum over z and a, and then using measurable near-minimisers, gives (9).

The finite-perimeter version should be proved by the same spacetime Gauss–Green argument applied to

$$\mathcal{E} := \{(\rho,x) : x \in E_\rho\} \subset [\rho_*,\rho_\delta] \times B_R,$$

whose perimeter is supplied by the coarea formula. This is a genuine limsup deformation theorem, not a lower-semicontinuity passage. It is the remaining load-bearing theorem.

7 Verdict

The state after this attack is sharper than before.

1. The same-centre Fusco–Julin input is closed, but the proof in the current preliminaries should be replaced. The correct source is Fusco–Julin's same oscillation centre plus their Poincare-type control of the symmetric difference.

2. The rescaled finite-perimeter metric theorem is still not proved by the manuscript's Reshetnyak/AGS passage; the inequality direction is wrong.

3. The bad-density kinetic obstruction is not intrinsic. It is an artifact of rescaling rough sets. For the unscaled moving-ball distance $q(\rho)$, nestedness gives a uniform Lipschitz bound and the bad-density set costs $O(\delta_T)$.

4. The remaining theorem is now the raw moving-ball first variation (9). It is local, good-level, and finite-perimeter; proving it with a direct spacetime limsup deformation would close the bounded Saint–Venant route without invoking BDV stability.

References

[1] N. Fusco and V. Julin, *A strong form of the quantitative isoperimetric inequality*, Calc. Var. Partial Differential Equations **50** (2014), 925–937, https://arxiv.org/abs/1111.4866.

[2] N. Fusco, *The quantitative isoperimetric inequality and related topics*, Bull. Math. Sci. **5** (2015), 517–607, https://link.springer.com/article/10.1007/s13373-015-0074-x.

[3] F. Maggi, *Sets of Finite Perimeter and Geometric Variational Problems*, Cambridge University Press, 2012.

[4] L. Ambrosio, N. Fusco, and D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford University Press, 2000.